Object Identification and Recognition (II)

Introduction to Computational and Biological Vision

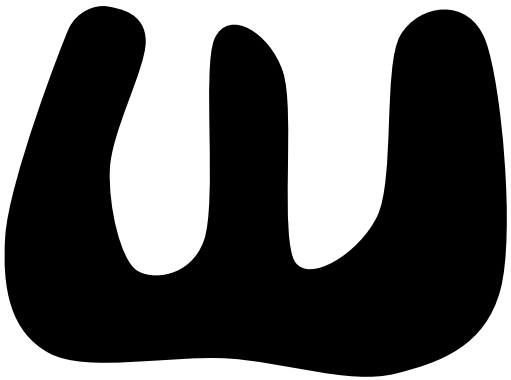
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

2D Regions and Shape Analysis

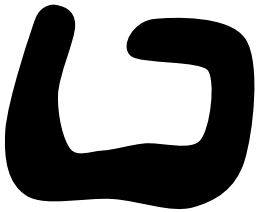
2D Shape representation



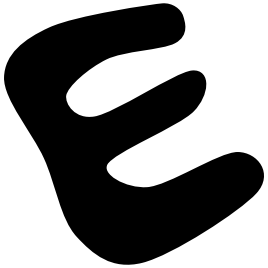
Coding and representation

$$X = (x_1, x_2, \ldots, x_n) \overset{?}{\approx}$$

$$Y^1 = (y_1^1, y_2^1, \ldots, y_n^1) \Leftarrow$$

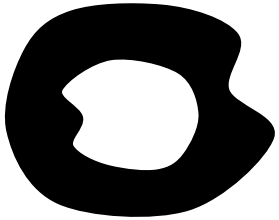


$$Y^2 = (y_1^2, y_2^2, \ldots, y_n^2) \Leftarrow$$



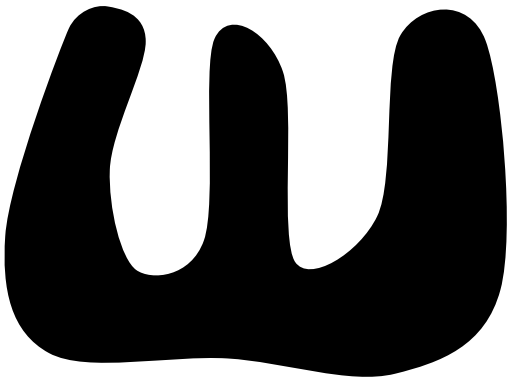
$$\vdots$$

$$Y^m = (y_1^m, y_2^m, \ldots, y_n^m) \Leftarrow$$



2D Regions and Shape Analysis

2D Shape representation: What can the features describe?



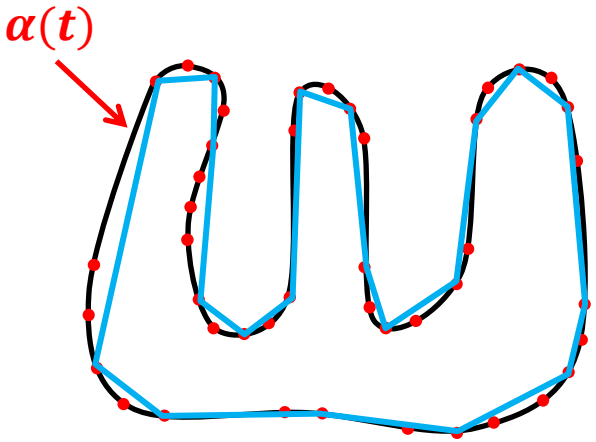
Coding and representation

$$X = (x_1, x_2, \ldots, x_n)$$

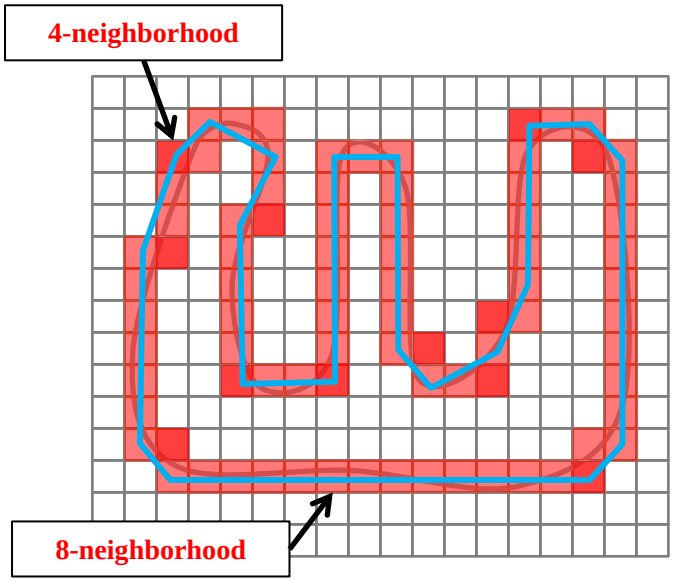
Representation	Contour (more compact)	Region (more explicit)

2D Regions and Shape Analysis

Contour representation



- Parametric curve (continuous)
- Point list (continuous and discrete)
- Chain code (discrete)
- Polygonal approximation (continuous and discrete)



NW	N	NE
W		E
SW	S	SE

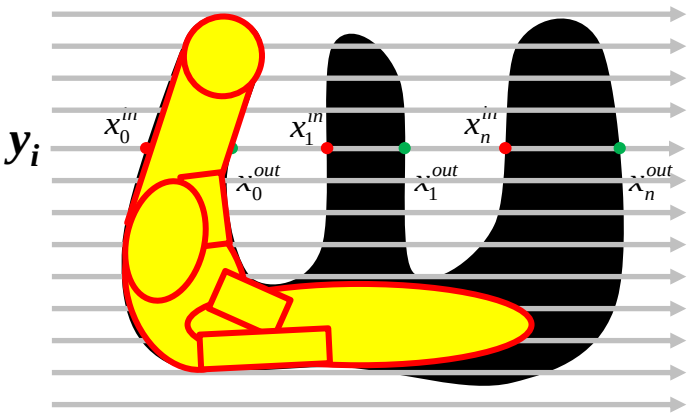
1,1,2,1,0,7,7,6,6,6,5,5,5,5,4,4,4,4,4,3,3,2,33,2,1,1,0,0,7,7,7

FL	F	FR
L		R
BL	B	BR

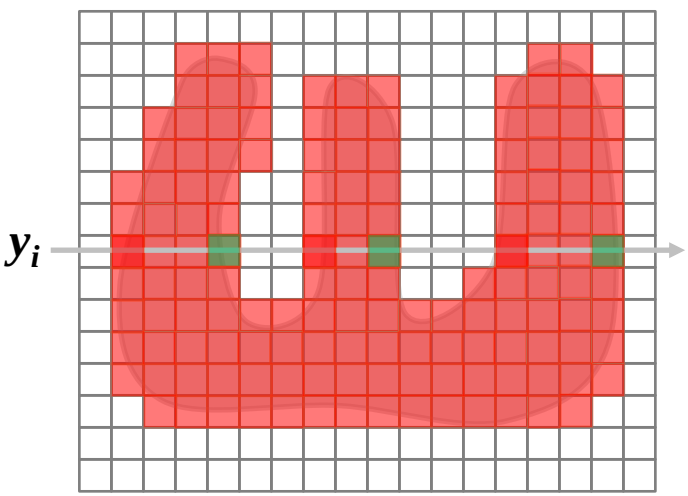
1,2,3,1,1,1,2,1,2,2,2,1,2,2,2,1,2,2,2,2,1,2,1,3,,2,1,1,2,1,2,1,2,2

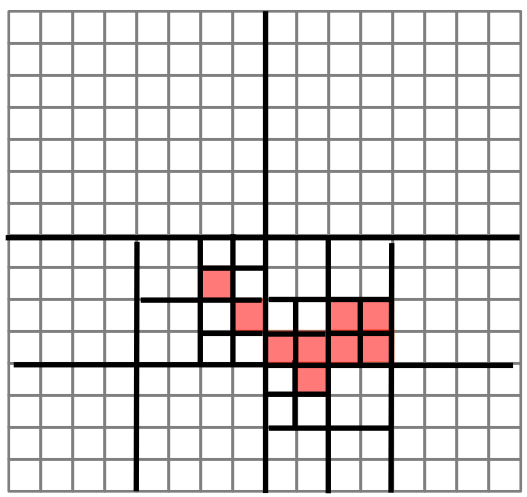
2D Regions and Shape Analysis

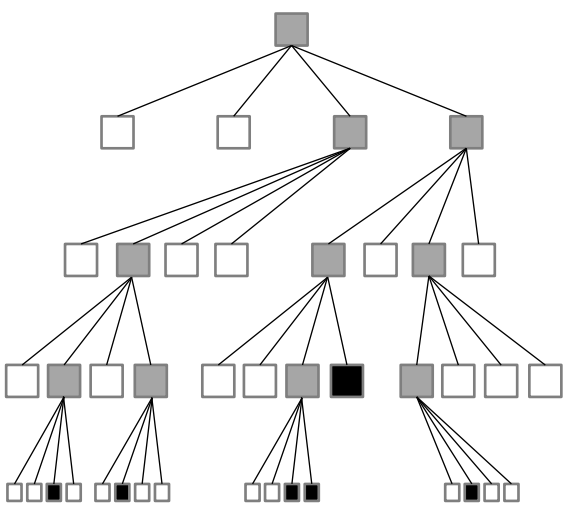
Area representation



- Explicit (discrete)
- Sweeping lines (constant/varying orientation)
- Standard regions
- Hierarchical representations (e.g., quad tree)

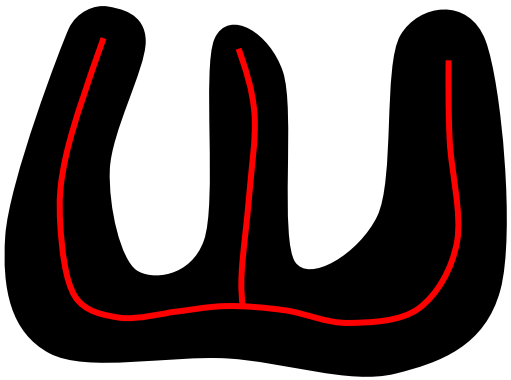




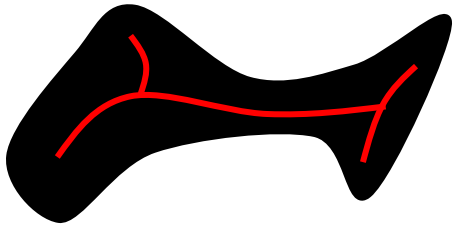


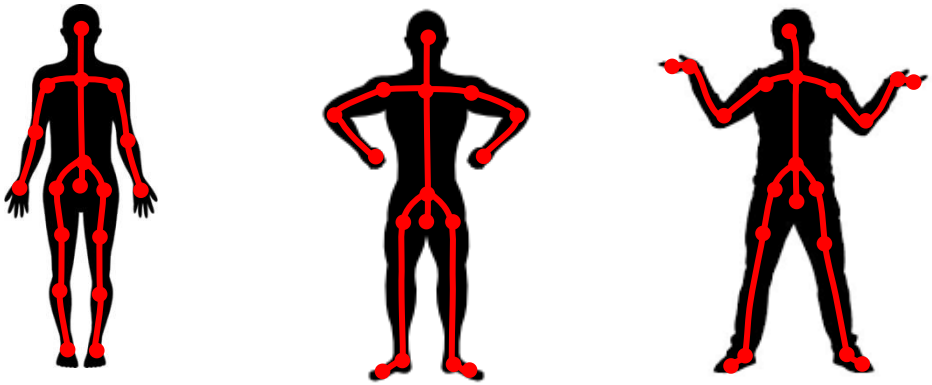
2D Regions and Shape Analysis

Skeleton representation



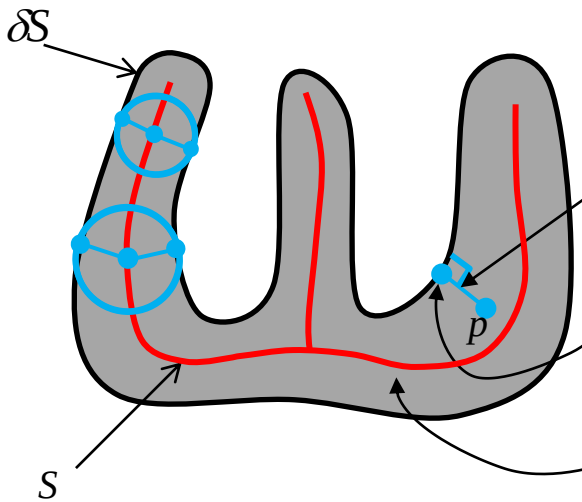
- 1D curve (or graph) representation for 2D regions
- Captures explicitly
    - shape geometry
    - general structure of the shape in terms of parts
    - general “strokes”
    - local symmetry





2D Regions and Shape Analysis

Skeleton representation via the medial axis



$$d(p,\delta S) = \min_{q \in \delta S}(p,q)$$

$$B(p,S) = \{q \in \delta S \mid d(p,q) = d(p,\delta S)\}$$

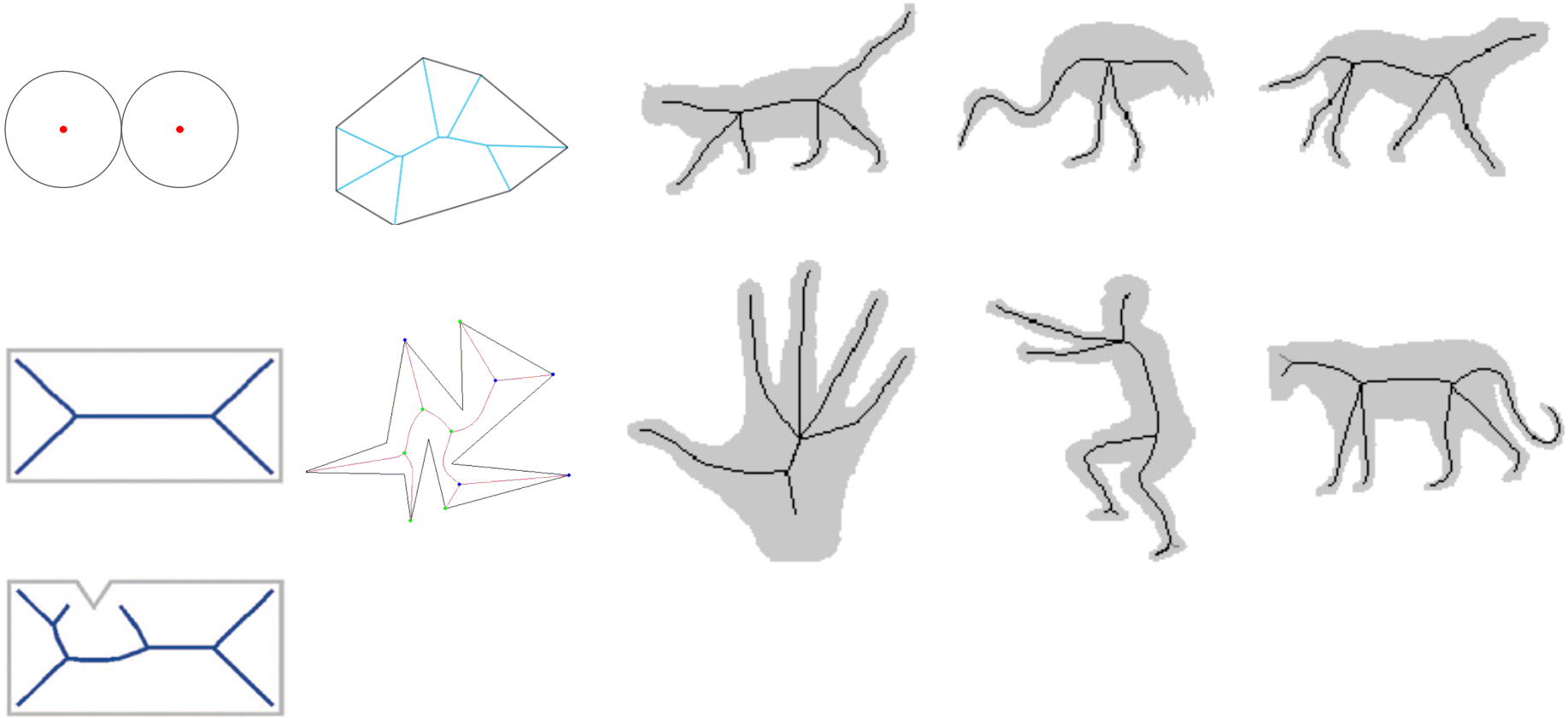
$$S^*(S) = \{p \in S \mid \|B(p,S)\| > 1\}$$

Medial axis: the set of all points having more than one closest point on the object's boundary.

Medial axis Transform: $S \Leftrightarrow MA(S) = \{(p, d(p,\delta S)) \mid p \in S^*(S)\}$

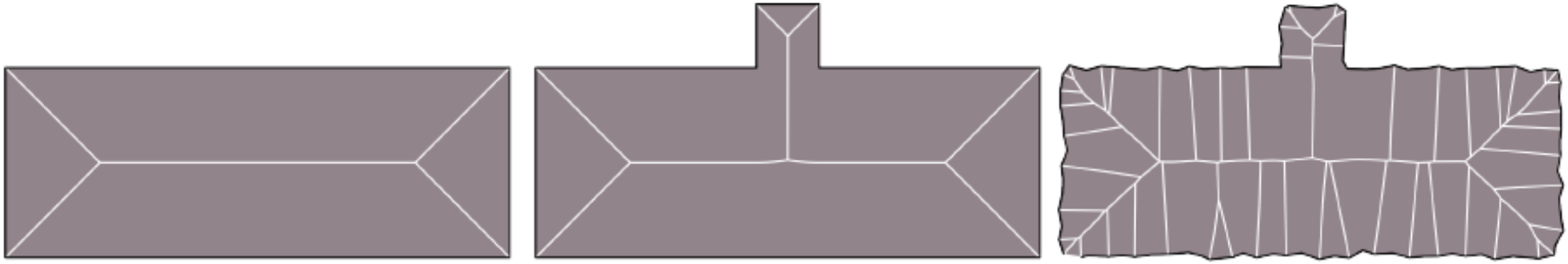
2D Regions and Shape Analysis

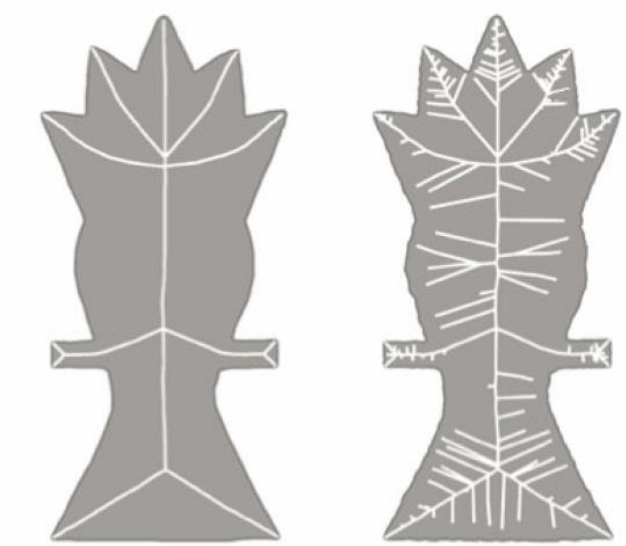
Skeleton representation via the medial axis



2D Regions and Shape Analysis

Skeleton representation via the medial axis





2D Regions and Shape Analysis

Shape invariants

Coding and representation

$$X = (x_1, x_2, \ldots, x_n)$$

$$\boldsymbol{x_i = 1 \Leftrightarrow S}$$ **is <your shape type of interest>**

$$\boldsymbol{f(X) = C \Leftrightarrow S}$$ **is <your shape type of interest>**

<u>Example (…perhaps the only obvious one)</u>

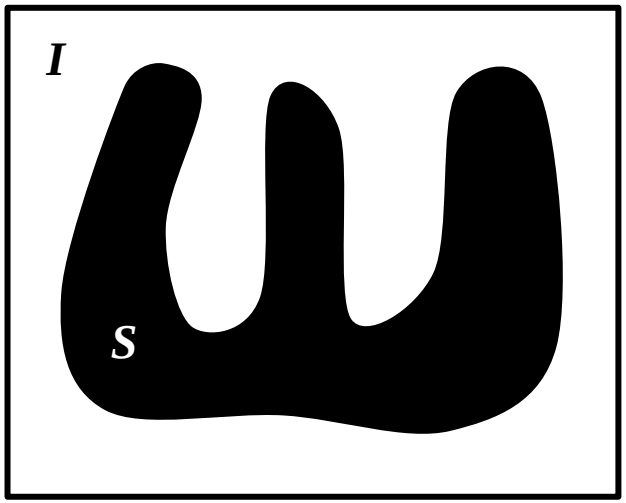
$$x_1 = Area(S)$$
$$x_2 = Perimeter(S)$$

$$x_3 = SF(S) = 4\pi \frac{x_1}{x_2^{\,2}}$$

$$S = \begin{cases} \text{Circle} & x_3 = 1 \\ \text{Non circle} & x_3 < 1 \end{cases}$$

2D Regions and Shape Analysis

Shape Moments (from region representation)



$$X = (x_1, x_2, \ldots, x_n)$$

$$h_S(x,y) = \begin{cases} 0 & (x,y) \notin S \\ 1 & (x,y) \in S \end{cases}$$

$$M_{pq} = \iint_I h_S(x,y) x^p y^q \, dx \, dy$$

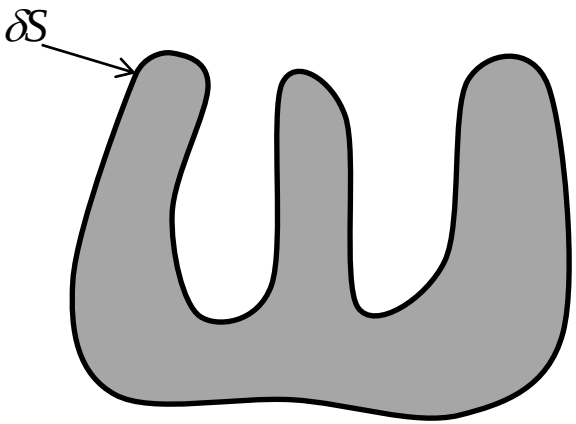
$$M_{pq} = \iint_S x^p y^q \, dx \, dy$$

$$M_{pq} = \sum_{(i,j) \in I} h_S(i,j) i^p j^q$$

$$M_{pq} = \sum_{(i,j) \in S} i^p j^q$$

2D Regions and Shape Analysis

Shape Moments (from contour representation)



$$\iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dxdy = \oint_{\delta D} Ldx + Mdy$$

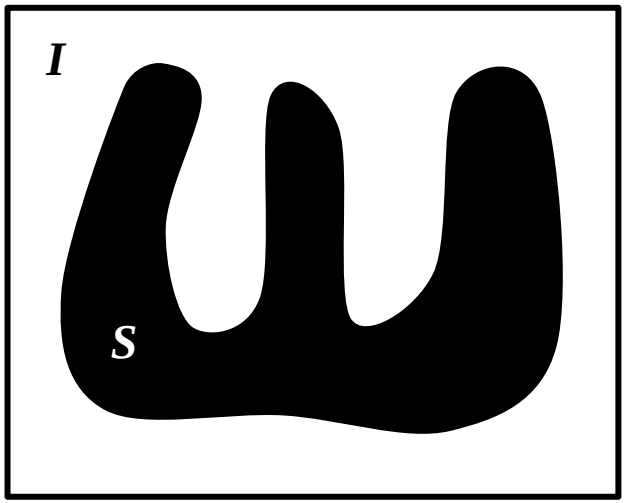
$$L(x, y) = 0$$

$$M(x, y) = \frac{1}{p+1} x^{p+1} y^q$$

$$M_{pq} = \iint_S x^p y^q dxdy = \iint_S \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dxdy = \oint_{\delta S} \frac{1}{p+1} x^{p+1} y^q dy$$

2D Regions and Shape Analysis

Shape Moments



$$h_S(x,y) = \begin{cases} 0 & (x,y) \notin S \\ 1 & (x,y) \in S \end{cases}$$

$$M_{pq} = \iint h_S(x,y)\, x^p y^q \, dxdy$$

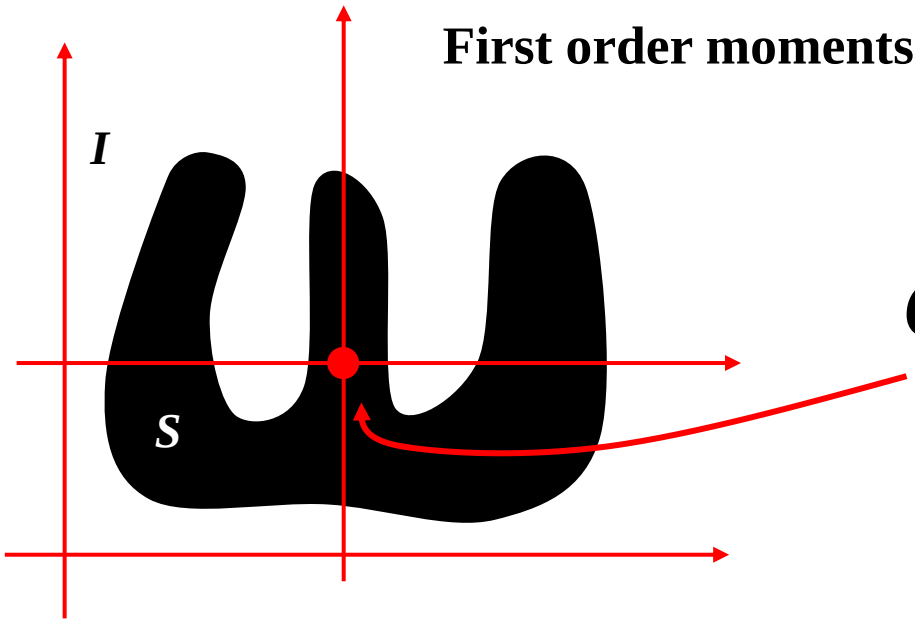
$$M_{00} = \iint_I h_S(x,y)\, dxdy$$

$$M_{10} = \iint_I x \cdot h_S(x,y)\, dxdy$$

$$M_{01} = \iint_I y \cdot h_S(x,y)\, dxdy$$

2D Regions and Shape Analysis

First order moments and central moments



$$C = \left(\frac{M_{10}}{M_{00}}, \frac{M_{01}}{M_{00}} \right)$$

$$M_{pq} = \iint_I h_S(x,y) x^p y^q \, dx dy$$

$$\overline{M}_{pq} = \iint_I h_S(x,y)(x-\bar{x})^p (y-\bar{y})^q \, dx dy$$

2D Regions and Shape Analysis

Second order central moments

$$\overline{M}_{20} = \iint_S (x - \bar{x})^2 \, dxdy$$

$$\overline{M}_{20} = \iint_S x^2 dxd \qquad - \bar{x}^2 \iint_S dxdy$$

$$\bar{x} = M_{10} / M_{00}$$

$$\overline{M}_{20} = M_{20} - 2\bar{x}M_{10} + \bar{x}^2 M_{00}$$

$$\overline{M}_{20} = M_{20} - \bar{x}^2 M_{00}$$

2D Regions and Shape Analysis

Second order moments

$$\overline{M}_{20} = M_{20} - \bar{x}^2 M_{00} \qquad \mu'_{20} = \overline{M}_{20} / \overline{M}_{00} = M_{20} / M_{00} - \bar{x}^2$$

$$\overline{M}_{11} = M_{11} - \bar{x}\bar{y} M_{00} \qquad \mu'_{11} = \overline{M}_{11} / \overline{M}_{00} = M_{11} / M_{00} - \bar{x}\bar{y}$$

$$\overline{M}_{02} = M_{02} - \bar{y}^2 M_{00} \qquad \mu'_{02} = \overline{M}_{02} / \overline{M}_{00} = M_{02} / M_{00} - \bar{y}^2$$

$$\tilde{\boldsymbol{C}} = \boldsymbol{COV}[\boldsymbol{S}] = \begin{bmatrix} \mu'_{20} & \mu'_{11} \\ \mu'_{11} & \mu'_{02} \end{bmatrix} = \boldsymbol{R} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \boldsymbol{R}^T$$

$$\tilde{\boldsymbol{C}}\boldsymbol{R} = \boldsymbol{R} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

2D Regions and Shape Analysis

Second order moments

$$\overline{M}_{20} = M_{20} - \bar{x}^2 M_{00} \qquad \mu'_{20} = \overline{M}_{20} / \overline{M}_{00} = M_{20} / M_{00} - \bar{x}^2$$

$$\overline{M}_{11} = M_{11} \qquad \mu'_{11} = \overline{M}_{11} / \overline{M}_{00} = M_{11} / M_{00} - \bar{x}\bar{y}$$

$$\overline{M}_{02} = M_{02} - \bar{y}^2 \qquad \mu'_{02} = \overline{M}_{02} / \overline{M}_{00} = M_{02} / M_{00} - \bar{y}^2$$

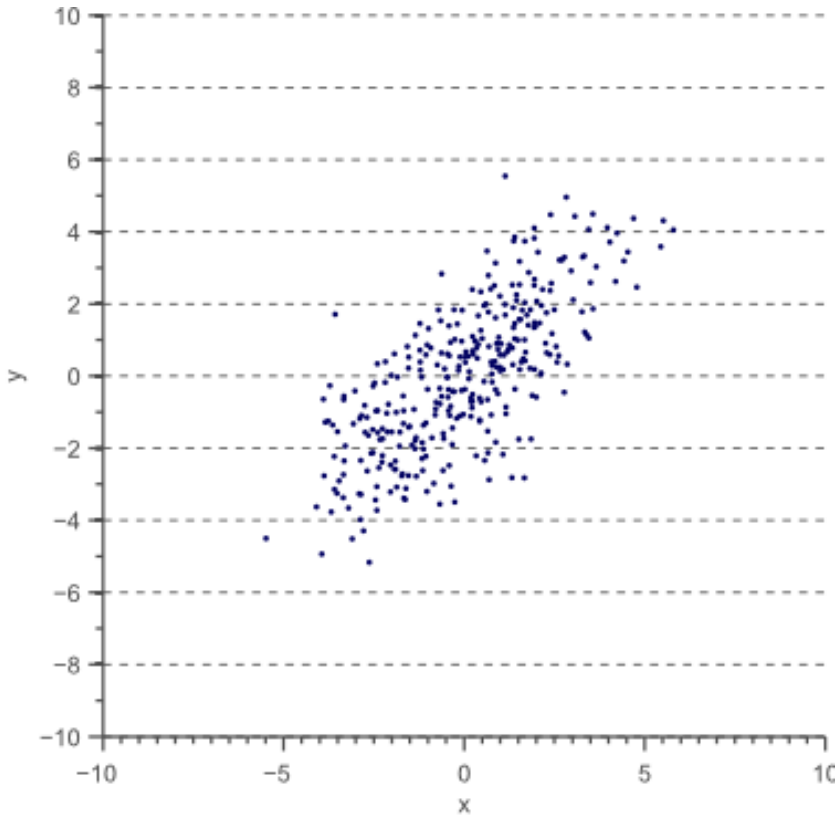
$$R = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$\tilde{C} = COV[S] = \begin{bmatrix} \mu'_{20} & \mu'_{11} \\ \mu'_{11} & \mu'_{02} \end{bmatrix} = R \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} R^T$$

$$\tilde{C}R = R \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

2D Regions and Shape Analysis

Reminder about variance and covariance

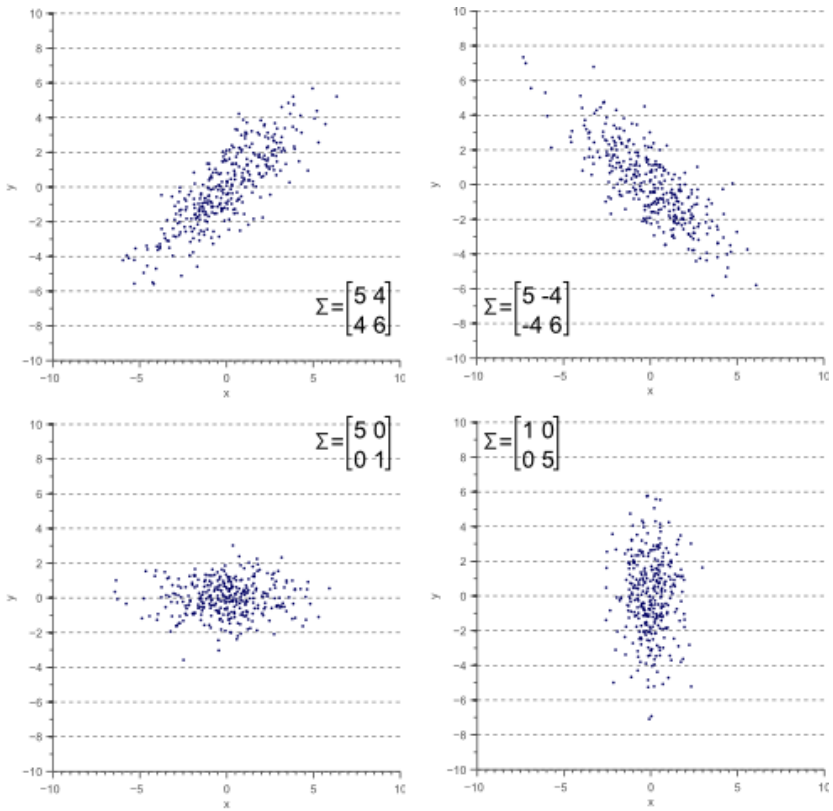


$$\sigma_x^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2 = \sigma(x,x) = \mathbb{E}[(x-\mathbb{E}(x))(x-\mathbb{E}(x))]$$

$$\sigma(x,y) = \mathbb{E}[(x-\mathbb{E}(x))(y-\mathbb{E}(y))]$$

2D Regions and Shape Analysis

Reminder about variance and covariance



$$\sigma_x^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2 = \sigma(x,x) = \mathbb{E}[(x-\mathbb{E}(x))(x-\mathbb{E}(x))]$$

$$\sigma(x,y) = \mathbb{E}[(x-\mathbb{E}(x))(y-\mathbb{E}(y))]$$

$$\Sigma = \begin{bmatrix} \sigma(x,x) & \sigma(x,y) \\ \sigma(y,x) & \sigma(y,y) \end{bmatrix}$$

2D Regions and Shape Analysis

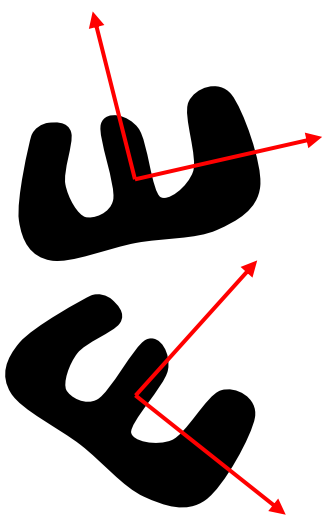
Second order moments

$$\begin{bmatrix} \mu'_{20} & \mu'_{11} \\ \mu'_{11} & \mu'_{02} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Shape orientation :

$$\theta = \frac{1}{2}\arctan\frac{2\mu'_{11}}{\mu'_{20} - \mu'_{02}}$$

$$\lambda_i = \frac{1}{2}\left(\mu'_{20} + \mu'_{02} \pm \sqrt{4\mu'^2_{11} + (\mu'_{20} - \mu'_{02})^2}\right)$$

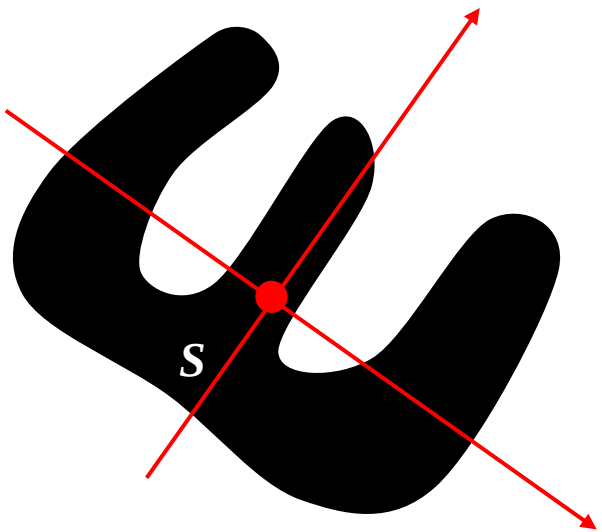


$$\begin{bmatrix} \mu'_{20} & \mu'_{11} \\ \mu'_{11} & \mu'_{02} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \bar{\mu}'_{20} & \bar{\mu}'_{11} \\ \bar{\mu}'_{11} & \bar{\mu}'_{02} \end{bmatrix} = \begin{bmatrix} \cos\bar{\theta} & \sin\bar{\theta} \\ -\sin\bar{\theta} & \cos\bar{\theta} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \cos\bar{\theta} & -\sin\bar{\theta} \\ \sin\bar{\theta} & \cos\bar{\theta} \end{bmatrix}$$

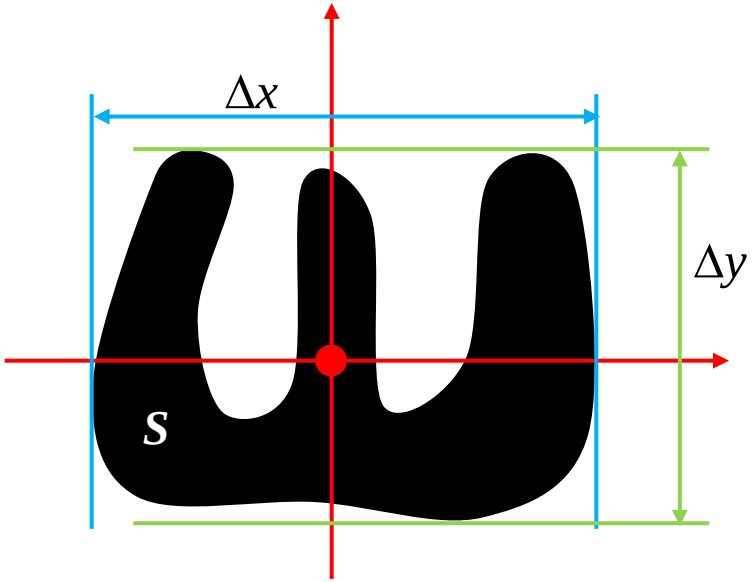
2D Regions and Shape Analysis

Shape eccentricity



$$\boldsymbol{e} = \sqrt{\mathbf{1} - \frac{\boldsymbol{\lambda}_{\mathbf{2}}}{\boldsymbol{\lambda}_{\mathbf{1}}}} \qquad \lambda_2 \geq \lambda_1$$

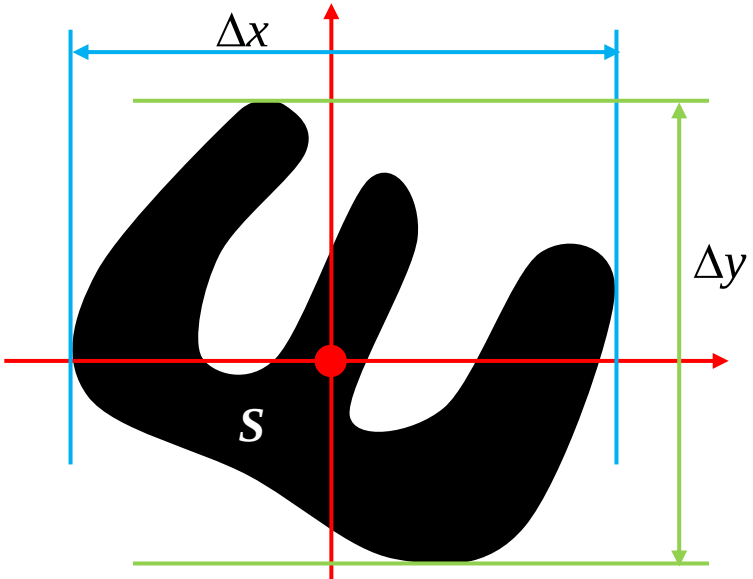
2D Regions and Shape Analysis



Aspect ratio

$$AR = \frac{\Delta y}{\Delta x}$$

$$AR(\theta) = \frac{\Delta y(\theta)}{\Delta x(\theta)}$$

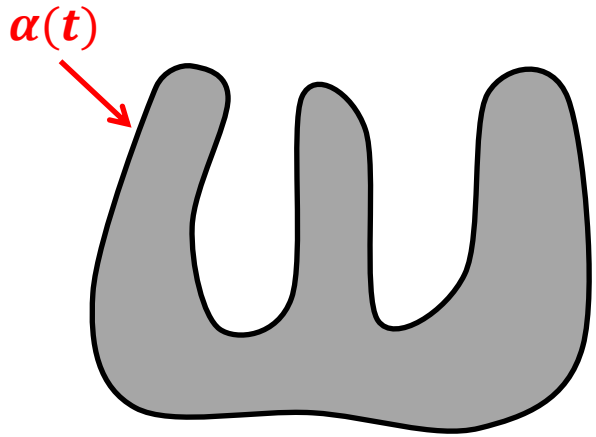


$$\overline{AR}(\theta) = \max\left\{AR(\theta) \mid \theta \in [0..2\pi]\right\}$$

…or alternatively $\overline{AR}(\theta) = \lambda_1 / \lambda_2$

2D Regions and Shape Analysis

Total boundary curvature



$$\alpha(s) \Rightarrow \kappa(s)$$

$$E = \int_0^l \kappa^2(s)ds$$

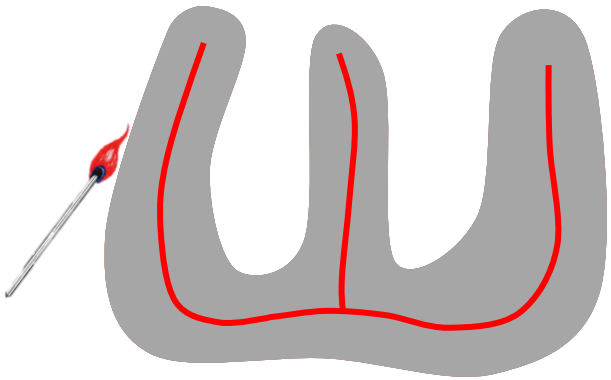
$$E[\text{circle of radius R}] = \int_0^{2\pi R} \frac{1}{R^2} ds = \frac{2\pi}{R}$$

Scale invariant

$$E = \frac{l}{4\pi^2} \cdot \int_0^l \kappa^2(s)ds$$

2D Regions and Shape Analysis

Computing skeletons



The naïve approach :

- For each interior point p of S

  Find the set $B(p,S)$ of closest boundary points

  If $||B(p,S)||>1$ mark as a skeleton point

Blum's Grassfire Transform :

- Set the boundary "on fire"
- Mark points where fire fronts meet

2D Regions and Shape Analysis

Skeletonization via Distance Transform

0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

Binary Image

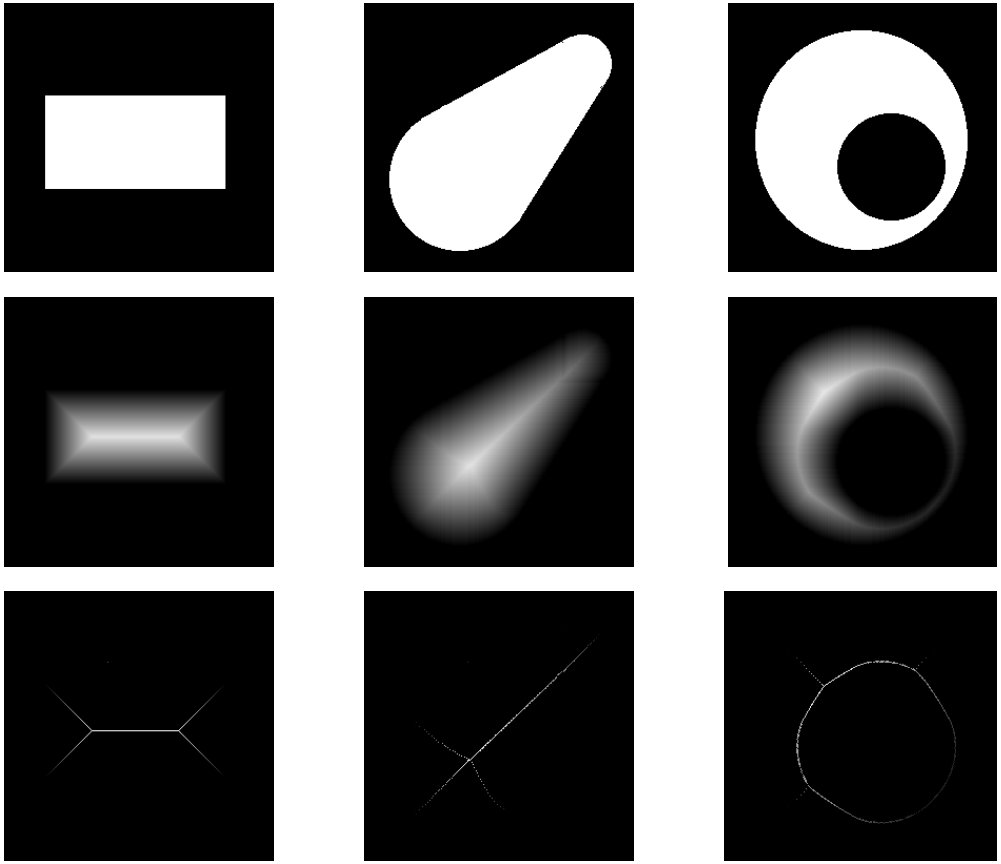
0	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	2	2	2	1	0
0	1	2	3	2	1	0
0	1	2	2	2	1	0
0	1	1	1	1	1	0
0	0	0	0	0	0	0

Distance transformation

$$
\begin{aligned}
&D(p) \leftarrow 0 \quad \forall p \in Image \\
&Front \leftarrow \partial S \\
&D(p) \leftarrow 1 \quad \forall p \in Front \\
&While\ Front \neq \emptyset \\
&\qquad Pick\ one\ pixel\ p \in Front \\
&\qquad Front \leftarrow Fron \setminus \{p\} \\
&\qquad For\ all\ q \in Nighborhood(p) \\
&\qquad\qquad D'(q) = D(p) + d(p,q) \\
&\qquad\qquad if\ D'(q) < D(q) \\
&\qquad\qquad\qquad D(p) \leftarrow D'(q) \\
&\qquad\qquad\qquad Front \leftarrow Front\ \cup \{q\} \\
&\qquad\qquad end \\
&\qquad end \\
&end
\end{aligned}
$$

2D Regions and Shape Analysis

Medial Axis Transform and Skeletonization via Distance Transform



Image/Shape

Distance Transform

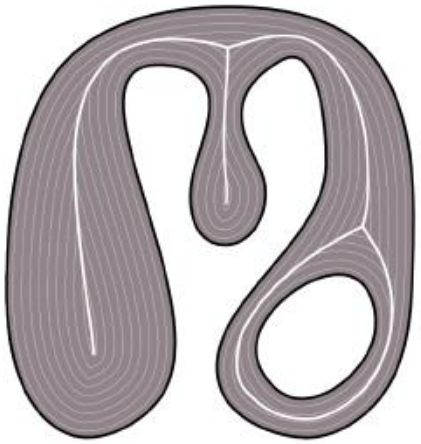
Ridges and singularities

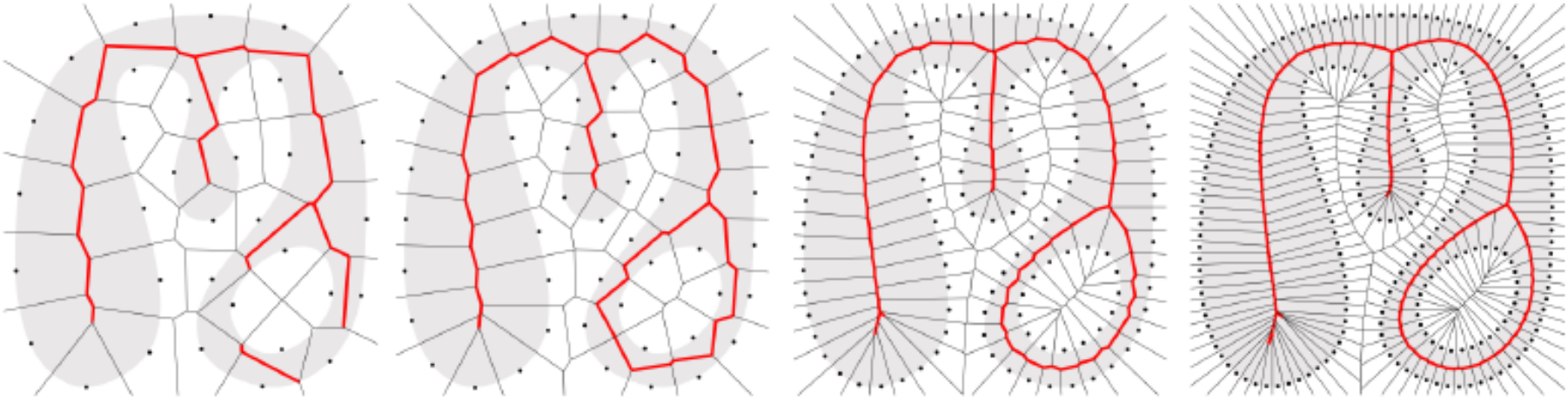
Medial Axis Transform

2D Regions and Shape Analysis

Voronoi Diagrams for skeletonization

- Sample the boundary of shape *S* densely enough.
- Compute the Voronoi diagram *V.*
- Take the subset of V that is fully contained in *S*.





[Tagliasacchi 2014]